

Factory Inspection Report for High/Low Voltage Substations

Project: JZP25JC032501 Export Guatemala Project Transformer Substation

Model: YBM-24/0.8-3300

Factory No: 2503645550

Date of production: July, 2025

1. Main components inside the cabinet:

Main component name	Production Plant	Factory number
☒ Transformer	JZPE POWER TRANSFORMER INC.	MY2506842
☒ Circuit-breaker	Schneider	12909902 07909902
☒ Low voltage frame circuit breaker	Schneider	07897071EN

2. Main technical parameters:

Parameter	Unit	Technical parameters	
Rated voltage on high and low voltage sides	kV	13.2/0.8	
Rated current on high and low voltage sides	A	144.3/2381.6	
Rated capacity	kVA	3300	
Insulation level	kV	Rated short-time power frequency withstand voltage: 38	
		Rated lightning impulse withstand voltage: 95	
High voltage tap voltage	V	1	13860
		2	13530
		3	13200
		4	12870
		5	12540
Transformer connection group label		YNd11	
Protection level of shell/fuel tank	IP	IP54/IP68	
Rated frequency	Hz	60	
Altitude	m	≤1200	
Power frequency withstand voltage correction factor	/	1.02	

3. Factory test:

3.1. Dielectric testing of the main circuit (8.2) :

Inspection items	Inspection content and standard requirements	Inspection results
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Dielectric testing of the main circuit	Conduct power frequency voltage testing on the high-voltage interconnection between high-voltage switchgear, control equipment, and power transformers. Apply voltage according to the system voltage in the table below, and there should be no breakdown or flashover phenomenon during the testing process;			No breakdown without flashover
	Rated voltage	Alternating and relative to each other	Fracture	
	7.2 kV	20 kV, 1min	23 kV, 1min	/
	12 kV	28 kV, 1min	32 kV, 1min	/
	17.5 kV	38 kV, 1min	45 kV, 1min	/
	24 kV	50 kV, 1min	60 kV, 1min	No breakdown or flashover
	36 kV	70 kV, 1min	80 kV, 1min	/

3.2. Auxiliary and control circuit testing (8.3.1、8.3.2、8.3.3、8.3.4)

Inspection items	Inspection content and standard requirements		Inspection results
Check the auxiliary and control circuits and verify compliance with the circuit diagram and wiring diagram	Check that the actuators, interlocking devices, locks, etc. are installed completely and operate reliably		Qualified
	Check if the components of the auxiliary and control circuits inside the casing are installed correctly		Qualified
	Check if the components of the auxiliary and control circuits inside the casing are installed correctly		Qualified
	Check if the wiring of wires and cables is correct		Qualified
	Check that conductors and cables will not be mechanically damaged due to sharp edges or proximity of heating elements or movement of moving parts		Qualified
	Check the identification of components and terminals, as well as the identification of cables and wiring (if applicable)		Qualified
	Simulate transformer oil level signal, gas signal, pressure signal, and oil temperature over temperature signal. Signal actions such as over temperature trip signal, high voltage cabinet load switch opening and closing signal, high voltage cabinet grounding switch opening and closing signal, high voltage isolation switch opening and closing signal, high voltage circuit breaker opening and closing signal, smoke sensor signal, box transformer door signal, etc. The signal display corresponding to the remote signaling switch is functioning properly.		Qualified
Mechanical operation test	All manually operated components, main switches, withdrawable functional units, etc. should be operated continuously in both forward and reverse directions for 5 cycles in an uncharged state, and should rotate flexibly		Qualified
	The mechanical interlocking between different components is reliable and meets the five prevention requirements		Qualified
Operation of the release device	Close the circuit 5 times at 85% of the rated control voltage, and it should close normally	Testing voltage : AC <u>187</u> V	Qualified
	Open the circuit 5 times with rated control voltage of 85%AC or 70% DC, and it should open normally	Testing voltage : AC <u>187</u> V	Qualified

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	Open and close 5 times at 110% rated control voltage, and should close normally	Testing voltage : AC <u>242</u> V	Qualified
	Operate continuously for 3 times with a rated control voltage of $\leq 30\%$, and do not open or close	Testing voltage : AC <u>66</u> V	Qualified
Electric shock protection verification	Under the condition of 30A (DC) current, the voltage drop from any possible grounding point in the prefabricated substation to the main grounding point of the substation is less than 3V		Qualified
Dielectric testing	Apply 2000V between the phases and relative ground of the main circuit of the low-voltage side device for 1 minute without breakdown or flashover		Qualified
	Connect the auxiliary circuit and control circuit on the high and low voltage sides to the ground (frame) and apply 1000V for 1 minute without breakdown or flashover		Qualified
Low voltage insulation resistance test	Using a 500V DC insulation measuring instrument for insulation measurement, if the insulation resistance between the circuit and the exposed conductive parts is at least 1000 Ω/V (for each circuit, the power supply voltage of these circuits is to ground), it is considered to have passed the test.		Qualified
	A-B ($M\Omega$)	B-C ($M\Omega$)	
	500	500	

3.3. Design and visual inspection (8.6)

Inspection items	Inspection content and standard requirements	Inspection results
External dimensions and installation dimensions	Compliant with technical documents	Qualified
Structural facilities, coating and cladding quality	Compliant with technical documents	Qualified
Inspection of component markings and nameplates inside the device.	The component markings, product model specifications, and nameplate markings meet the requirements	Qualified
Visually inspect that the fixing screws of the outer box are securely installed.	The fixing screws of the outer box are securely installed, and the screws of the door hinge are marked with lines	Qualified

3.4. Other functional tests (8.101)

Inspection items	Inspection content and standard requirements	Inspection results
Device function test	Each functional component can operate and indicate normally when subjected to rated voltage or current.	Qualified
Mechanical operation of the door	The opening, closing, and interlocking of the box door should be flexible and reliable. (The door panel and instrument panel can be opened flexibly without any obstruction, with an opening angle of $\geq 90^\circ$ and a positioning device)	Qualified

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Inspection of operational channels	The operation channel is suitable for any operation and maintenance, with a width of $\geq 800\text{mm}$	Qualified
Inspection of Transformer Temperature and Liquid Level	The value displayed on the transformer thermometer is consistent with the temperature at the sampling point of the transformer, and the value displayed on the oil level gauge is within the marked range	Qualified
Inspection of voltage indication	The low voltage side voltmeter indicates correctly; The high-voltage side live display is displaying normally	Qualified
Connection of grounding device	The cross-section of the main grounding conductor (copper) is not less than 30mm^2 ; The screw connecting the grounding conductor should be $\leq 12\text{mm}$ and have a grounding label	Qualified
Operation of Transformer Tap changer	Flexible and reliable operation	Qualified

Note: in the above inspection results indicates the absence of this inspection item.

Conclusion: This product has undergone factory inspection and meets the requirements of IEC 62271-202 and IEC 60076-1-2011 standards, and is allowed to leave the factory.

Inspection: ZhouXiang

Audit: WangLiwen



JIEZOU POWER

TEST REPORT

Product Name: Power transformer

Type: SL-3300/13.2

Product Code: 1RZ.720.C458

Factory number: MY Y2506842

Manufacture date: 2025.06

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Test Report

1. Technical parameters

Type	SL-3300/13.2	Vector Group	YNd11	Factory Number	MYY2506842
Rated Capacity	3300KVA	Phase	3	Product Code	1RZ.720.C458
Service Condition	Outdoor	Rated Frequency	60 Hz	Type of Cooling	ONAN
Rated voltage and rated current	Taping	HV		LV	
		Voltage (V)	Current (A)	Voltage (V)	Current (A)
	1	13860	137.5	800	2381.6
	2	13530	140.8		
	3	13200	144.3		
	4	12870	148.0		
	5	12540	151.9		

2.Winding resistance measurement

Temperature:30.9℃ Relative humidity:70%

HV	Taping	OA (Ω)	OB (Ω)	OC (Ω)
	1	0.17530	0.17563	0.17506
	2	0.17159	0.17209	0.17164
	3	0.16972	0.17008	0.16948
	4	0.16526	0.16555	0.16484
	5	0.15992	0.16032	0.15992
LV		ab (Ω)	bc (Ω)	ca (Ω)
		0.0007395	0.0007547	0.0007558

3. Voltage ratio measurement and joint group label verification

Measured voltage ratio deviation (%)	Taping	AB/ab	BC/bc	CA/ca	Vector group
	1	0.01	0.01	0.02	YNd11
	2	-0.11	-0.11	-0.11	
	3	0.10	0.09	0.09	
	4	-0.03	-0.03	-0.03	
	5	0.18	0.18	0.18	

4. Transformer core and clamp insulation inspection

Temperature: 30.1°C Relative humidity: 70%

Instrument	Test voltage (V)	Core insulation to ground (MΩ)	Iron core connected to the grounding plate	Remarks
Digital Megger	2500	≥500	The iron core and the clamp are connected to the ground	Before oiling

5. Insulation resistance measurement

Temperature : 30.1°C Relative humidity: 70%

Using the instrument	Measuring site	Test voltage (V)	Measured insulation resistance (MΩ)
Digital megger	HV - LV+E	2500	4200
	LV - HV+E	2500	4170
	HV+LV-E	2500	4230

6. Applied voltage test

Temperature : 31.5°C Relative humidity: 70%

Pressurized part	Test voltage(kV)	Test time(s)	Result
HV - LV+E	38	60	Pass
LV - HV+E	3	60	Pass
Insulation of auxiliary wiring	2.5	60	Pass

7. Induction withstand voltage test

Temperature : 31.5°C Relative humidity: 75%

Applied voltage	Inductive voltage	Induction multiple	Frequency(Hz)	Test time(s)	Result
Low pressure(kV)	High pressure(kV)				
1.6	26.4	2	150	48	Pass

8. No-load current and no-load loss measurement

Temperature : 31.5°C Relative humidity: 75%

Apply voltage	No-load loss (W)	No-load current(%)	Reference capacity (kVA)
90%Ur	2317	0.11	3300
100%Ur	2968	0.18	3300
110%Ur	4077	0.42	3300

9. Short circuit impedance and load loss measurement

Temperature :32.0℃

Relative humidity:73%

Winding	Taping	75℃ load loss (W)	75℃ Short circuit impedance(%)
HV-LV	3	22080	5.54

10.Measurement of dielectric loss factor of insulation system

Temperature:34.0℃

Relative humidity:70%

Assembly	HV-LV+E	LV-HV+E	HV+LV-E
Applied voltage (kV)	10	0.8	0.8
$\tan \delta$ (%)	0.30	0.86	0.78
Cx (nF)	12.00	73.51	85.02

11.Rated working efficiency

η	99.241%
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12. Seal test

Experiment method	Applying pressure (kPa)	Duration (h)	Result
Air pressure method	50	24	No leakage, no damage

**13.Insulating fluid test
pre-test**

Type of oil		Breakdown voltage (kV)		Dielectric loss factor (90℃)		Slight water (mg/L)	
TVAI-30℃		66.2		0.00149		9.6	
Colour spectrum (μL/L)							
H2	CO	CO2	CH4	C2H4	C2H6	C2H2	Total hydrocarbon
0.59	0.26	281.55	0.81	0.11	0.00	0.00	0.92

post-test

Type of oil		Breakdown voltage (kV)		Dielectric loss factor (90 ℃)		Slight water (mg/L)	
TVAI-30℃		65.8		0.00152		9.4	
Colour spectrum (μL/L)							
H2	CO	CO2	CH4	C2H4	C2H6	C2H2	Total hydrocarbon
0.67	0.32	296.62	0.84	0.13	0	0	0.97

This product has been tested, meet the international standards IEC 60076-1 provisions, the quality of parts meet the drawing and technical conditions, approved to factory.

Statement:

1. The test report is invalid without the seal of the test center. 2. The test report is invalid without the signature of the approver. 3. Any objection to the test report shall be raised promptly to the test center upon receipt of the test report. 4. The test results are valid only for the sample. 5. The report shall not be reproduced without the written approval of the testing center.

Compile: Liang jinyun

Test: Xiao guojiang Xian runkang

Approved:  Date: 2015.7.15

End line
